Introduction to Beamer

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Introduction

What is Beamer?

- Beamer is a LaTeX document class used to create presentations.
- It allows us to create structured slides.
- It supports equations, tables, figures and more.

Document and Page Styles

Document Class

- Defines the overall structure of a document.
- Examples:
  - article
  - report
  - book

Page Style

- Controls the appearance of headers and footers.
- Examples: plain, empty, headings and fancy.

Mathematical Equations

Beamer also supports mathematical expressions.

$$E = mc^2$$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Advantages

- Mathematical expressions are easy to format.
- Equations can be displayed clearly on slides.

Using Blocks to Highlight Information

Beamer provides blocks that separate key information from the rest of the slide.

Regular Block

Use this style for definitions, explanations, or important notes.

Important Notice

Use an alert block sparingly when a point needs special attention.

Example

A short example helps connect a concept with a practical use.

Arranging Content in Columns

Columns place related material side by side instead of in one long list.

Text and Lists

- Explain an idea.
- Give supporting points.
- Keep each point concise.

Examples and Visuals

- Show a small table.
- Add a diagram.
- Include an equation.

Use columns when two pieces of information are easier to understand together.

Revealing Content Step by Step

Overlays let a presenter reveal selected items gradually instead of showing everything at once.

A list with step-by-step appearance

- First, introduce the main idea.

Useful overlay commands

- `\pause` pauses before showing the next content.
- `\only<2->{...}` displays material from overlay 2 onward.
- `\visible<2->{...}` reserves space before material appears.

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Revealing Content Step by Step

Overlays let a presenter reveal selected items gradually instead of showing everything at once.

A list with step-by-step appearance

- First, introduce the main idea.
- Next, explain a supporting detail.
- Finally, show the conclusion.

Useful overlay commands

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Themes and Visual Customization

Beamer themes control the overall appearance of a presentation, including colors, navigation elements, and slide headings.

Selecting a theme

```
\usetheme{Madrid}
\usecolortheme{seahorse}
```

Design suggestions

- Choose a theme that suits the purpose of the presentation.
- Use a small set of matching colors.
- Keep headings and body text visually distinct.
- Avoid putting too much information on one slide.

Tips for an Effective Presentation

A good slide deck supports the speaker rather than replacing the speaker.

- **Keep slides focused:** present one main idea on each slide.
- **Use readable text:** choose font sizes visible from a distance.
- **Prefer concise points:** explain details verbally instead of filling slides with paragraphs.
- **Use visuals carefully:** add diagrams, tables, or images when they clarify the topic.
- **Check consistency:** keep colors, headings, and spacing uniform.
- **Rehearse:** practise moving through the slides before presenting.

Thank You!

Questions?

Key Takeaways

- Beamer is used to create presentations using LaTeX.
- Each slide is created using the frame environment.
- We can include text, equations, tables and figures.